

Question 2

2(b)

```
clc, clearvars
function nll = negloglik(params, prices, choices)
    a1 = params(1);
    a2 = params(2);
    b = params(3);
    % V: n x 3
    V = [a1 + b*prices(:,1), a2 + b*prices(:,2), 0 + b*prices(:,3)];
    maxV = max(V,[],2); % numerical stability
    expV = exp(V - maxV); % subtract max per row
    denom = sum(expV,2);
    P = expV ./ denom;
    % form log-likelihood: sum_i log(P(i, choice_i))
    idx = sub2ind(size(P), (1:size(P,1))', choices);
    logP_chosen = log(P(idx));
    % negative log-likelihood
    nll = - sum(logP_chosen);
end

data = readtable('cola.xlsx', 'VariableNamingRule','preserve');
ids= unique(data.id);
n= numel(ids);
y = zeros(n,1); % chosen brand (1=Pepsi, 2=7-Up, 3=Coke)
p1 = zeros(n,1); % price of Pepsi
p2 = zeros(n,1); % price of 7-Up
p3 = zeros(n,1); % price of Coke

for ii= 1:n
    idx= data.id==ids(ii);
    sub= data(idx,:);
    p1(ii)= sub.price(1);
    p2(ii)= sub.price(2);
    p3(ii)= sub.price(3);
    y(ii)= find(sub.choice==1);
end
price_mat= [p1 p2 p3];

%MLE
init= [0;0;-1];
opts= optimoptions("fminunc", "Algorithm","quasi-newton","Display","iter",...
    'MaxFunctionEvaluations',1e5, 'MaxIterations',5e4);
fun= @(p) negloglik(p, price_mat, y);
[phat, fval, exitflag, output, grad, hess]= fminunc(fun, init, opts);
```

Iteration	Func-count	f(x)	Step-size	First-order optimality
0	4	1882.82		81
1	8	1843.15	0.0123493	68.2
2	12	1829.57	1	60.9
3	20	1824.86	0.315155	10.9
4	24	1824.57	1	0.82

5	28	1824.56	1	0.12
6	32	1824.56	1	0.0033
7	36	1824.56	1	6.1e-05

Computing finite-difference Hessian using objective function.

Local minimum found.

Optimization completed because the size of the gradient is less than the value of the optimality tolerance.

<stopping criteria details>

```
hessian_used= hess;
method_used= 'fminunc';

a1_hat= phat(1);
a2_hat= phat(2);
b_hat= phat(3);

covmat= inv(hessian_used);
se= sqrt(diag(covmat));
tstats= phat./se;

fprintf('\n--- MLE results (method: %s) ---\n', method_used);
```

--- MLE results (method: fminunc) ---

```
rowNames = {'alpha1 (Pepsi)', 'alpha2 (7-Up)', 'beta (price)'};
results = table(phat, se, tstats, 'RowNames', rowNames, 'VariableNames',
{'Estimate', 'StdErr', 'tStat'});
disp(results);
```

	Estimate	StdErr	tStat
alpha1 (Pepsi)	0.28317	0.06238	4.5394
alpha2 (7-Up)	0.10383	0.062469	1.6622
beta (price)	-2.2964	0.13766	-16.681

2(c)

```
p_test = [1.00, 1.25, 1.10];
V_test = [a1_hat + b_hat*p_test(1), a2_hat + b_hat*p_test(2), 0 +
b_hat*p_test(3)];
expV= exp(V_test);
P_test= expV./sum(expV);
disp('Predicted choice probabilities at prices [Pepsi, 7-Up, Coke] = [1.00,
1.25, 1.10]:');
```

Predicted choice probabilities at prices [Pepsi, 7-Up, Coke] = [1.00, 1.25, 1.10]:

```
disp(array2table(P_test, 'VariableNames', {'Pepsi', '7Up', 'Coke'}));
```

Pepsi	7Up	Coke
_____	_____	_____

0.48319 0.22746 0.28934

2(d)

```
delta = 0.10;  
P = P_test; % baseline vector [P1 P2 P3]  
fprintf('\nMarginal effects for +$%.2f increase in each alternative price  
(approx):\n', delta);
```

Marginal effects for +\$0.10 increase in each alternative price (approx):

```
for j = 1:3  
    fprintf('When price of alt %d increases by $%.2f:\n', j, delta);  
    for k = 1:3  
        if k == j  
            dPk = b_hat * P(j) * (1 - P(j)) * delta;  
        else  
            dPk = - b_hat * P(j) * P(k) * delta;  
        end  
        fprintf('  Change in P%d: %g\n', k, dPk);  
    end  
end
```

When price of alt 1 increases by \$0.10:

Change in P1: -0.0573443

Change in P2: 0.0252392

Change in P3: 0.0321052

When price of alt 2 increases by \$0.10:

Change in P1: 0.0252392

Change in P2: -0.0403527

Change in P3: 0.0151136

When price of alt 3 increases by \$0.10:

Change in P1: 0.0321052

Change in P2: 0.0151136

Change in P3: -0.0472187